

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

9MA0/31

Mathematics

Advanced

PAPER 31: Statistics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/



Pearson

1. George throws a ball at a target 15 times.

Each time George throws the ball, the probability of the ball hitting the target is 0.48

The random variable X represents the number of times George hits the target in 15 throws.

(a) Find

(i) $P(X = 3)$

(ii) $P(X \geq 5)$

(3)

George now throws the ball at the target 250 times.

- (b) Use a normal approximation to calculate the probability that he will hit the target more than 110 times.

(3)

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Question 1 continued

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(Total for Question 1 is 6 marks)



2. A manufacturer uses a machine to make metal rods.

The length of a metal rod, L cm, is normally distributed with

- a mean of 8 cm
- a standard deviation of x cm

Given that the proportion of metal rods less than 7.902 cm in length is 2.5%

(a) show that $x = 0.05$ to 2 decimal places.

(2)

(b) Calculate the proportion of metal rods that are between 7.94 cm and 8.09 cm in length.

(1)

The **cost** of producing a single metal rod is 20p

A metal rod

- where $L < 7.94$ is **sold** for scrap for 5p
- where $7.94 \leq L \leq 8.09$ is **sold** for 50p
- where $L > 8.09$ is shortened for an extra **cost** of 10p and then **sold** for 50p

(c) Calculate the expected profit per 500 of the metal rods.
Give your answer to the nearest pound.

(5)

The same manufacturer makes metal hinges in large batches.

The hinges each have a probability of 0.015 of having a fault.

A random sample of 200 hinges is taken from each batch and the batch is accepted if fewer than 6 hinges are faulty.

The manufacturer's aim is for 95% of batches to be accepted.

(d) Explain whether the manufacturer is likely to achieve its aim.

(4)



Question 2 continued

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Question 2 continued

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Question 2 continued

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(Total for Question 2 is 12 marks)



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(3)

(2)

(1)

Question 3 continued

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(Total for Question 3 is 7 marks)



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Question 4 continued

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(Total for Question 4 is 6 marks)



5. A company has 1825 employees.

The employees are classified as professional, skilled or elementary.

The following table shows

- the number of employees in each classification
- the two areas, A or B , where the employees live

	A	B
Professional	740	380
Skilled	275	90
Elementary	260	80

An employee is chosen at random.

Find the probability that this employee

(a) is skilled,

(1)

(b) lives in area B and is not a professional.

(1)

Some classifications of employees are more likely to work from home.

- 65% of professional employees in both area A and area B work from home
- 40% of skilled employees in both area A and area B work from home
- 5% of elementary employees in both area A and area B work from home
- Event F is that the employee is a professional
- Event H is that the employee works from home
- Event R is that the employee is from area A

(c) Using this information, complete the Venn diagram on the opposite page.

(4)

(d) Find $P(R' \cap F)$

(1)

(e) Find $P([H \cup R]')$

(1)

(f) Find $P(F | H)$

(2)

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Question 5 continued

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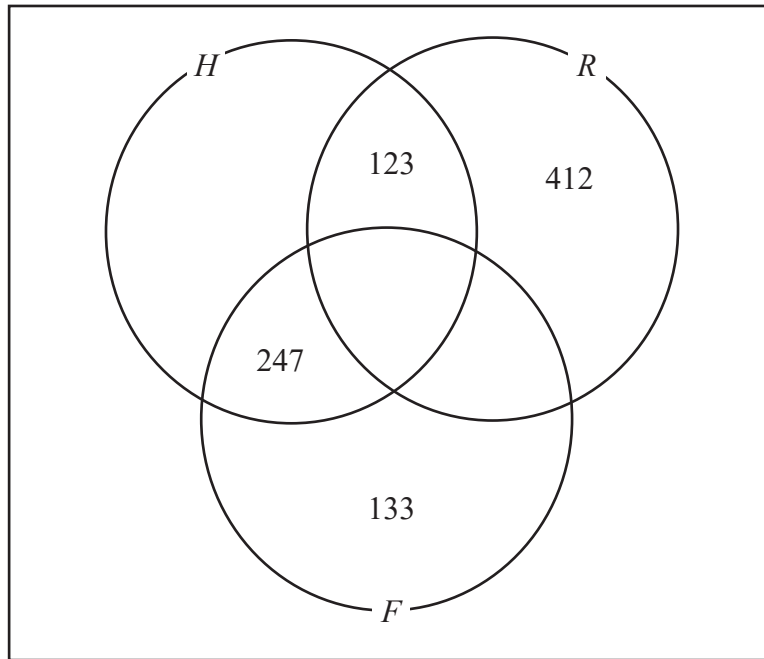
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Question 5 continued

Only use this diagram if you need to redraw your Venn diagram.



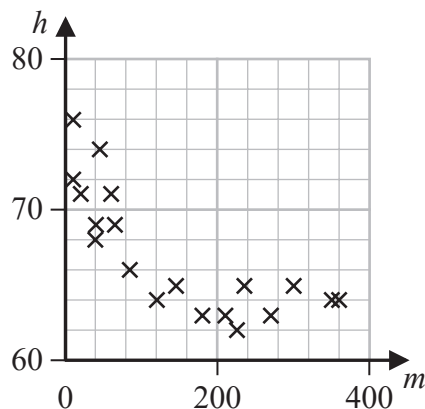
(Total for Question 5 is 10 marks)



6. Anna is investigating the relationship between exercise and resting heart rate. She takes a random sample of 19 people in her year at school and records for each person

- their resting heart rate, h beats per minute
- the number of minutes, m , spent exercising each week

Her results are shown on the scatter diagram.



- (a) Interpret the nature of the relationship between h and m

(1)

Anna codes the data using the formulae

$$x = \log_{10} m$$

$$y = \log_{10} h$$

The product moment correlation coefficient between x and y is -0.897

- (b) Test whether or not there is significant evidence of a negative correlation between x and y

You should

- state your hypotheses clearly
- use a 5% level of significance
- state the critical value used

(3)

The equation of the line of best fit of y on x is

$$y = -0.05x + 1.92$$

- (c) Use the equation of the line of best fit of y on x to find a model for h on m in the form

$$h = am^k$$

where a and k are constants to be found.

(5)

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Question 6 continued

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Question 6 continued

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Question 6 continued

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Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 9 marks)

TOTAL FOR STATISTICS IS 50 MARKS



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Pearson Edexcel Level 3 GCEPaper
reference**9MA0/32****Mathematics****Advanced****PAPER 32: Mechanics****You must have:**

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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Q:1/1/1/1/

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1. [In this question, position vectors are given relative to a fixed origin.]

At time t seconds, where $t > 0$, a particle P has velocity $\mathbf{v} \text{ m s}^{-1}$ where

$$\mathbf{v} = 3t^2\mathbf{i} - 6t^{\frac{1}{2}}\mathbf{j}$$

- (a) Find the speed of P at time $t = 2$ seconds. (2)

- (b) Find an expression, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, for the acceleration of P at time t seconds, where $t > 0$ (2)

At time $t = 4$ seconds, the position vector of P is $(\mathbf{i} - 4\mathbf{j}) \text{ m}$.

- (c) Find the position vector of P at time $t = 1$ second. (4)

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Question 1 continued

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(Total for Question 1 is 8 marks)



2.

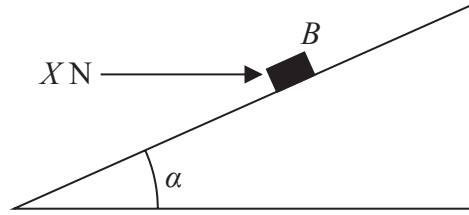


Figure 1

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A small block B of mass 5 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 1.

The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

The block B is modelled as a particle.

The magnitude of the normal reaction of the plane on B is 68.6 N .

Using the model,

(a) (i) find the magnitude of the frictional force acting on B , (3)

(ii) state the direction of the frictional force acting on B . (1)

The horizontal force of magnitude X newtons is now removed and B moves down the plane.

Given that the coefficient of friction between B and the plane is 0.5

(b) find the acceleration of B down the plane. (6)

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Question 2 continued

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Question 2 continued

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(Total for Question 2 is 10 marks)



3.

[In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 4 kg is at rest at the point A on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (4\mathbf{i} - \mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (\lambda\mathbf{i} + \mu\mathbf{j})\text{ N}$, where λ and μ are constants, are applied to P

Given that P moves in the direction of the vector $(3\mathbf{i} + \mathbf{j})$

(a) show that

$$\lambda - 3\mu + 7 = 0 \quad (4)$$

At time $t = 4$ seconds, P passes through the point B .

Given that $\lambda = 2$

(b) find the length of AB . (5)

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Question 3 continued

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 9 marks)



4.

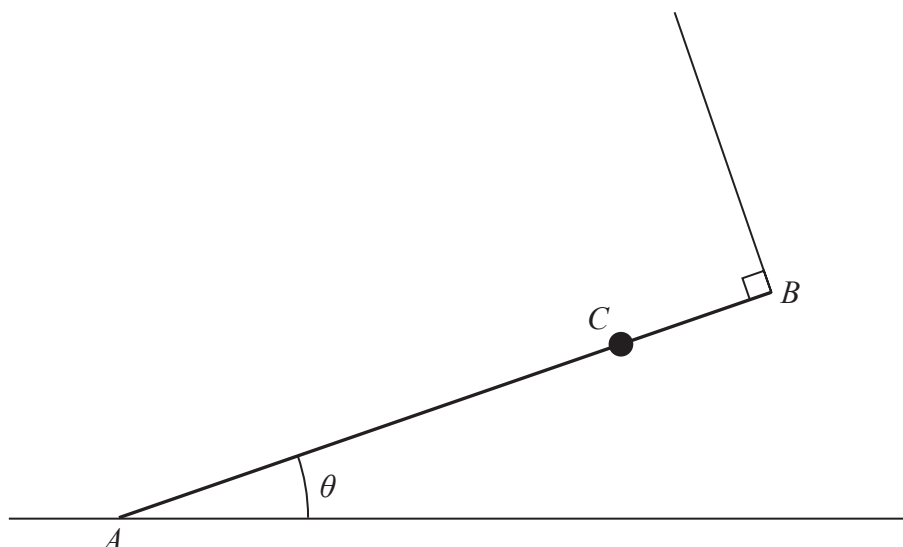


Figure 2

A uniform rod AB has mass M and length $2a$

A particle of mass $2M$ is attached to the rod at the point C , where $AC = 1.5a$

The rod rests with its end A on rough horizontal ground.

The rod is held in equilibrium at an angle θ to the ground by a light string that is attached to the end B of the rod.

The string is perpendicular to the rod, as shown in Figure 2.

- (a) Explain why the frictional force acting on the rod at A acts horizontally to the right on the diagram.

(1)

The tension in the string is T

- (b) Show that $T = 2Mg \cos \theta$

(3)

Given that $\cos \theta = \frac{3}{5}$

- (c) show that the magnitude of the vertical force exerted by the ground on the rod at A is $\frac{57Mg}{25}$

(3)

The coefficient of friction between the rod and the ground is μ

Given that the rod is in limiting equilibrium,

- (d) show that $\mu = \frac{8}{19}$

(4)

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Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 11 marks)



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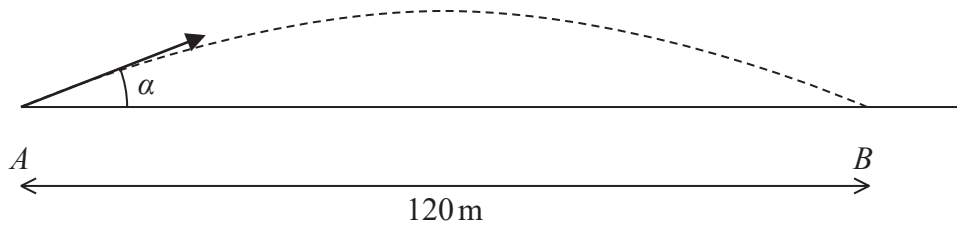


Figure 3

A golf ball is at rest at the point A on horizontal ground.

The ball is hit and initially moves at an angle α to the ground.

The ball first hits the ground at the point B , where $AB = 120\text{ m}$, as shown in Figure 3.

The motion of the ball is modelled as that of a particle, moving freely under gravity, whose initial speed is $U\text{ m s}^{-1}$

Using this model,

(a) show that $U^2 \sin \alpha \cos \alpha = 588$ (6)

The ball reaches a maximum height of 10 m above the ground.

(b) Show that $U^2 = 1960$ (4)

In a refinement to the model, the effect of air resistance is included.

The motion of the ball, from A to B , is now modelled as that of a particle whose initial speed is $V\text{ m s}^{-1}$

This refined model is used to calculate a value for V

(c) State which is greater, U or V , giving a reason for your answer. (1)

(d) State one further refinement to the model that would make the model more realistic. (1)

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Question 5 continued

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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 12 marks)

TOTAL FOR MECHANICS IS 50 MARKS

